Utility/Privacy Trade-off through the lens of Optimal Transport

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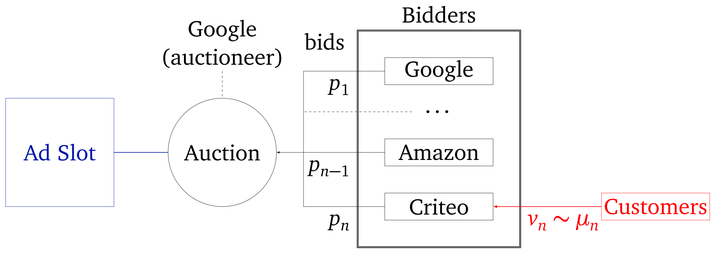
Summary

- Utility/Privacy trade-off motivated by economic mechanisms
- General minimization problem with regularization
- Convex problem + finite solution under mild assumptions
- Related to OT and Sinkhorn loss
- Simulation for repeated auctions

Motivation: repeated auctions

Auction for ad slot with private value v.
Bid $p \implies$ auctioneer infers μ, distribution of v.
Adjusted reserve price: auctioneer revenue $\nearrow$ and bidder utility $\searrow$ (long term).



Online advertisement auction framework

Goal: earn short term utility **and** hide μ for long term utility.

Toy example

Private type k drawn from prior $p_0 \in \mathcal{P}(\mathcal{Y})$
Utility loss $x^\top y_k \to$ objective: $\min_{x\in\mathcal{X}} x^\top y_k$
Randomization play $x \sim \mu_k \in \mathcal{P}(\mathcal{X})$
Adversary updates posterior p_x from Bayes rule
Privacy loss $\mathrm{KL}(p_x, p_0)$ given action x

New utility/privacy framework

Previous frameworks: minimize loss with privacy constraint [2]

Our objective: $\min_{\mu_1,\dots,\mu_K} \sum_{k=1}^K p_0(k)\mathbb{E}_{x\sim\mu_k}\left[x^\top y_k + \lambda \mathrm{KL}(p_x, p_0)\right]$

- regularized version $\to$ economic value of information
- true utility/privacy trade-off
- λ value of information

General formulation

Utility loss $c(x,y)$ for action x and type y
Privacy loss $\mathbb{E}_x\left[D(p_x, p_0)\right]$
Variable joint distribution $\gamma \in \mathcal{P}(\mathcal{X}\times\mathcal{Y})$ with marginal constraint

$$\inf_{\substack{\gamma\in\mathcal{P}(\mathcal{X}\times\mathcal{Y})\\ \pi_{2\#}\gamma = p_0}} \int_{\mathcal{X}\times\mathcal{Y}} \left[c(x,y) + \lambda\, D(p_x, p_0)\right] \mathrm{d}\gamma(x,y). \qquad \text{(OPT)}$$

Theoretical results

1. $D \in f$-divergences (e.g. KL or TV) $\implies$ (OPT) convex minimization
2. type space $\mathcal{Y}$ finite $\implies$ optimal γ with finite support (known size)

Relation with Optimal Transport

Sinkhorn loss $\mathrm{OT}_{c,\lambda}(\mu,\nu) = \min_{\gamma\in\Pi(\mu,\nu)} \int c(x,y)\mathrm{d}\gamma(x,y) + \lambda \int \log\left(\frac{\mathrm{d}\gamma(x,y)}{\mathrm{d}\mu(x)\mathrm{d}\nu(y)}\right)\mathrm{d}\gamma(x,y)$
$\to$ quickly approximate OT distances [1]

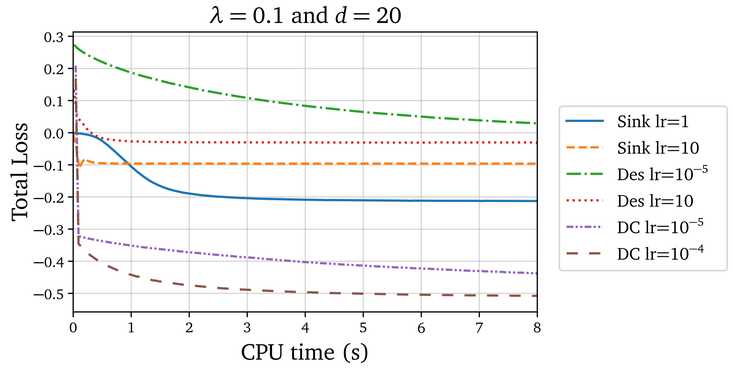
$D = \mathrm{KL} \implies$ (OPT) equivalent to $\inf_{\mu\in\mathcal{P}(\mathcal{X})} \mathrm{OT}_{c,\lambda}(\mu, p_0)$ (S−OPT)

- bridges privacy & optimal transport theories
- better comprehension of problem structure (lower dim.)
- finite $\mathcal{Y}$: look for $\mu = \sum_{k=1}^n \alpha_k \delta_{x_k}$

Experiment 1: toy example

3 optimization schemes:

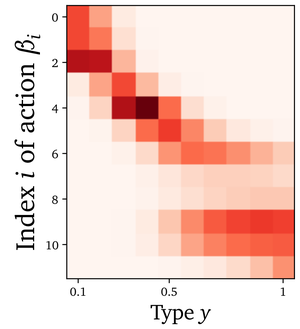
Des GD on (OPT) for $(\gamma, x) \in \mathbb{R}_+^{K\times n} \times \mathcal{X}^n$
Sink GD on (S-OPT) for $(\alpha, x) \in \mathbb{R}_+^{n} \times \mathcal{X}^n$
DC Difference of convex algorithm (linear utility only)



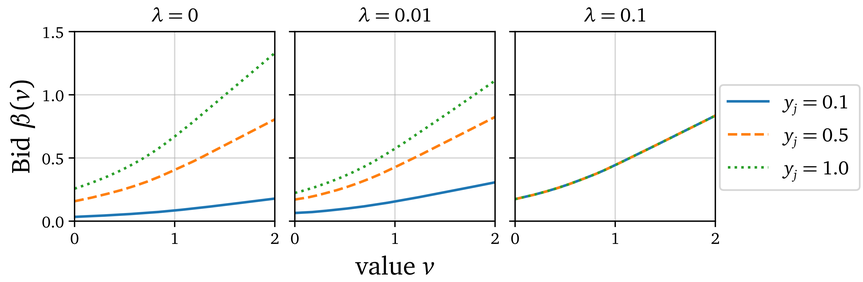
Comparing optimization schemes

Second experiment: repeated auctions

Type item value $v \sim \mu_{y_j} = \mathrm{Exp}(1/y_j)$ for a single round
Action bid $\beta_i(v/y_j)$ with mapping $\beta_i \in \mathbb{R}_+^{\mathbb{R}_+}$ (1 layer NN)
Adversary observe distribution $\rho_i = \beta_{i\#}\mu_1$
Utility known for lazy second price auction [3]



Heat map of joint distribution γ



Evolution of bidding strategy ($\lambda = 0.01$)

- randomize types over close values
- actions scale with type: less and less when $\lambda \nearrow$
- reveal more for small $y \to$ heterogeneous value of information

References

[1] Cuturi, M. (2013). Sinkhorn distances: Lightspeed computation of optimal transport. In *NIPS*.

[2] Eilat, R., Eliaz, K., and Mu, X. (2019). Optimal Privacy-Constrained Mechanisms. Technical report, C.E.P.R. Discussion Papers.

[3] Nedelec, T., Karoui, N. E., and Perchet, V. (2019). Learning to bid in revenue-maximizing auctions. In *International Conference on Machine Learning*, pages 4781–4789.